



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fourth Semester – Spring 2024

Paper: Statistics-IV

Course Code: STAT-203

Roll No.018...

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

1. What are basic principle of experimental design.

2. Fit Multiple regression $Y=a+bx_1+cx_2$

To following data

$$\sum Y = 89 \quad \sum x_1 = 30$$

$$\sum x_1^2 = 52 \quad \sum x_1 Y = 238$$

$$\sum x_2^2 = 582$$

$$\sum x_1 x_2 = 351 \quad \sum x_1 Y = 619$$

$$\sum x_2 Y = 1007 \quad n =$$

3. Write down the layout of Latin square design with 5 treatments.

4. Three varieties A,B,C of a crop are tested in CRD (complete randomize design) Analyse the experimental yield and state your conclusion.

A	B	C
813	647	713
C	B	A
814	759	795
A	C	C
795	652	598

5. Differentiate between simple and partial correlation.

6. Find the least value of r in a sample of 18 pairs from a bivariate normal population that is significant at 5% level

Answer the following questions

(3x10=30)

Q2. 492 candidates for scientific posts gave particulars of their university degrees and their hobbies. Discuss the association between two criteria of classification the degree and hobbies.

Categories	Maths	Chemistry	Physics
Music	24	83	17
Crafts work	11	62	28
Reading	32	121	34
Darma	10	26	44

Q3. Perform the ANOVA on the data and analyse the treatment means using LSD test with a $\alpha=0.05$

Block	Treatment					
	1	2	3	4	5	6
1	1	3	6	4	3	2
2	1	4	4	8	5	1
3	3	6	7	8	4	3
4	2	3	2	3	2	1

Q4. Find the linear regression equation from the following data

$$\sum x = 725$$

$$\sum Y = 1011$$

$$\sum x^2 = 44,475$$

$$\sum Y^2 = 85905$$

$$\sum XY = 61685$$

Assuming normality test the hypothesis

$$H_0: \alpha = 32$$

against

$$H_1: \alpha = 32$$

$$H_0: \beta = 0$$

against

$$H_1: \beta = 0$$

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED****Q.1. Answer the following short questions.****(6x5=30)**

1. Explain the in detail why it is not good statistical procedure to perform several t-test on pairs of mean. When several means are to be compared suggest the alternative procedure.
2. In a linear regression problem the following results are obtained
 $Y = 12.13 - 0.1608x$
 $\sum(Y - \hat{Y})^2 = 1160$ $\sum(x - \bar{x})^2 = 1300$ $n = 24$
 Find 95% confidence interval for α & β
3. What is coefficient of determination in multiple regression. What it explain.
4. Test whether there is any association between the results of preliminary and those of the final test.

		Preliminary Test	
Categories		Pass	Fail
Final	Pass	605	135
	Fail	195	65

5. Differentiate between partial and multiple correlation.
6. Find the simple correlation coefficient r_{12} , r_{13} and r_{23} for the following information
 $b_{12} = -0.1$ $b_{21} = -0.4$ $b_{13} = 0.27$
 $b_{31} = 0.6$ $b_{23} = 0.67$ $b_{32} = 0.38$

Answer the following questions**(3x10=30)**

Q2. Following is the plan of a field layout for the testing four varieties A, B, C and D of wheat in each 5 blocks. The plot yields in pound are also indicated there is

Block				
1	2	3	4	5
D 29.3	B 33.0	D 29.8	B 36.8	D 28.8
B 33.3	A 34.0	A 34.3	A 35.0	C 35.8
C 30.8	C 34.3	B 36.3	D 28.0	B 34.5
A 32.3	D 26.0	C 35.3	C 32.3	A 36.5

Perform an analysis of variance and state your conclusions.

Q3. Given the following data

X_1	1	4	1	3	2	4
X_2	1	8	3	5	6	10
X_3	2	8	1	7	4	6

- (a) Find the least square regression line where x_1 dependent variable and x_2 and x_3 are independent variable.
- (b) Calculate the coefficient of multiple determination and multiple correlation and interpret the result

Q4. Records taken of the number of male and female births 800 families having four children are as follows.

No of Male berth	0	1	2	3	4
Families	32	178	290	236	64

Test whether the data is consistent with hypothesis that the binominal law holds and that chance of a male birth is equal to that of female birth that is $p = q = 0.5$.